

Grouped Survey of Attention Mechanisms (Revised 2026)

Purpose

This revision follows the **grouped style with explicit formulas** that you requested. It also sharpens the taxonomy of attention mechanisms,

- positional-aware variants,
- sparse / approximate variants,
- KV-sharing and inference-efficient variants,
- domain-specific variants,
- and adjacent long-sequence alternatives that are **not literally attention**.

The key principle is:

$$\text{Attn}(Q, K, V) = \text{N}(\text{S}(Q, K) + B_{pos} + M_{struct}) V$$

where:

- $\text{S}(Q, K)$ is the similarity rule,
- B_{pos} is a positional bias or geometric term,
- M_{struct} is a structural mask or sparsity pattern,
- $\text{N}(\cdot)$ is a normalization / selection operator.

This decomposition makes it easier to classify modern attention research.

1. Dot-Product Based Attention Family

These mechanisms derive directly from similarity between query and key vectors.

1.1 Scaled Dot-Product Attention (Transformer)

$$A(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

Key idea

The factor $1/\sqrt{d_k}$ stabilizes the variance of logits and prevents softmax saturation when the key dimension is large.

Complexity

- time: $O(n^2 d)$
- memory: $O(n^2)$

Status

Foundational and mainstream.

1.2 Multi-Head Attention (MHA)

For head i :

$$\text{head}_i = \text{softmax}\left(\frac{QW_i^Q(KW_i^K)^\top}{\sqrt{d_k}}\right) V W_i^V$$

$$\text{MHA}(X) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O$$

Key idea

Different heads learn different interaction subspaces.

Benefit

Improves expressiveness compared with a single attention map.

Status

Canonical Transformer design.

1.3 Multi-Query Attention (MQA)

All query heads share one key projection and one value projection:

$$\text{head}_i = \text{softmax}\left(\frac{QW_i^Q(KW^K)^\top}{\sqrt{d_k}}\right) V W^V$$

Key idea

Reduce KV-cache size during autoregressive decoding.

Benefit

Much lower memory bandwidth cost at inference.

Tradeoff

Less flexible than full MHA.

Status

Very important in fast decoding systems.

1.4 Grouped-Query Attention (GQA)

Let query heads be partitioned into groups $g(i)$:

$$\text{head}_i = \text{softmax}\left(\frac{QW_i^Q(KW_{g(i)}^K)^\top}{\sqrt{d_k}}\right) V W_{g(i)}^V$$

Key idea

Interpolate between MHA and MQA.

Benefit

Preserves most of MHA quality while approaching MQA efficiency.

Status

Mainstream in modern decoder-only LLMs.

1.5 Talking-Heads Attention

A simplified form is:

$$A = \text{softmax}(W_L L W_R)$$

where

$$L = \frac{QK^\top}{\sqrt{d_k}}$$

Key idea

Mix logits or probabilities across heads before and/or after softmax.

Benefit

Allows interaction among heads rather than treating each head independently.

Status

Interesting refinement, not as central as GQA/MQA.

1.6 Cosine Attention

$$s_{ij} = \frac{q_i^\top k_j}{\|q_i\| \|k_j\|}$$

$$\alpha_{ij} = \text{softmax}(s_{ij})$$

Key idea

Normalize vector norms so attention depends on angular similarity.

Use

Memory networks, retrieval-style models, and norm-sensitive settings.

Status

Niche but conceptually clean.

1.7 Temperature Attention

$$A(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\tau}\right) V$$

Key idea

The temperature τ controls concentration.

- small τ : sharper attention
- large τ : smoother attention

Note

In practice, temperature may be fixed, learned, head-specific, or fused with scaling.

1.8 Differential Attention

A simplified form is:

$$A(Q, K, V) = [\text{softmax}(L^{(1)}) - \lambda \text{softmax}(L^{(2)})]V$$

Key idea

Subtract one attention map from another to suppress noisy or irrelevant mass.

Motivation

Address attention over-allocation to irrelevant context.

Status

Emerging 2024--2025 direction.

2. Additive / Energy-Based Attention Family

Attention is defined through an energy function rather than a pure dot product.

2.1 Bahdanau Attention

$$e_{ij} = v^\top \tanh(W_q q_i + W_k k_j)$$

$$\alpha_{ij} = \text{softmax}(e_{ij})$$

Key idea

A small neural network computes compatibility.

Importance

Historically central in seq2seq models before Transformers.

Status

Historical foundation.

2.2 General Energy Attention

$$e_{ij} = f_{\theta}(q_i, k_j)$$

where f_{θ} can be an MLP, tensor module, or other neural scoring function.

Key idea

Dot-product attention is one special case of energy-based attention.

2.3 Neural Tensor Attention

One form is:

$$e_{ij} = q_i^{\top} W^{[1:r]} k_j + U^{\top} [q_i; k_j] + b$$

where $W^{[1:r]}$ is a 3-way tensor.

Key idea

Capture richer pairwise interactions than linear or bilinear scoring.

Use

Relation modeling, matching, knowledge reasoning.

Status

Expressive but computationally heavier.

2.4 Hopfield Attention (Modern Hopfield Networks)

Energy:

$$E(x) = -\log \sum_i \exp(x^{\top} \xi_i)$$

Update:

$$x' = \frac{\sum_i \xi_i \exp(x^{\top} \xi_i)}{\sum_i \exp(x^{\top} \xi_i)}$$

Key idea

Associative memory retrieval has the same mathematical form as softmax attention.

Importance

Provides a strong theoretical bridge between attention and modern energy-based memory.

Status

Highly important conceptually.

3. Bilinear Attention Family

These generalize dot-product attention using learned bilinear forms.

3.1 Bilinear Attention

$$s_{ij} = q_i^\top W k_j$$

Key idea

Replace the identity metric with a learned matrix W .

Benefit

Greater flexibility than pure dot product.

3.2 Low-Rank Bilinear Attention

A factorized form is:

$$s_{ij} = q_i^\top U V^\top k_j$$

or equivalently:

$$s_{ij} = (U^\top q_i)^\top (V^\top k_j)$$

Key idea

Reduce bilinear parameter cost using low-rank factorization.

3.3 Bilinear Attention Networks (BAN)

BAN-style feature interaction often uses bilinear pooling between modalities:

$$z = \text{pool}(U^T x \odot V^T y)$$

Use

Vision-language tasks such as VQA.

Status

Domain-specific, not a core generic LLM attention mechanism.

3.4 Compact Bilinear Attention

Tensor-sketch approximation:

$$z = \text{FFT}^{-1}(\text{FFT}(a) \odot \text{FFT}(b))$$

Key idea

Approximate high-dimensional bilinear interactions efficiently.

4. Kernelized Attention Family

These rewrite or approximate attention through kernel feature maps.

A common linearized form is:

$$A(Q, K, V) = \frac{\phi(Q)(\phi(K)^T V)}{\phi(Q)(\phi(K)^T \mathbf{1})}$$

Key idea

Avoid explicit construction of the $n \times n$ attention matrix.

4.1 Linear Transformer

Typical positive feature map:

$$\phi(x) = \text{ELU}(x) + 1$$

Complexity

Often reduced from quadratic token-token interaction to linear-in-length accumulation.

Note

Actual complexity depends on implementation details and dimensions.

4.2 Performer (FAVOR+)

A random feature form is:

$$\phi_{\omega}(x) = \exp(\omega^{\top} x - \frac{\|x\|^2}{2})$$

Key idea

Approximate the softmax kernel using random features.

Importance

One of the best-known efficient attention families.

4.3 Gaussian Attention

$$s_{ij} = \exp(-\frac{\|q_i - k_j\|_2^2}{2\sigma^2})$$

Key idea

Attention behaves like an RBF kernel.

4.4 Laplacian Kernel Attention

$$s_{ij} = \exp(-\frac{\|q_i - k_j\|_1}{\sigma})$$

Key idea

Use an L_1 -based kernel for sharper locality.

4.5 cosFormer / Related Linearized Variants

In this family, the goal is not only approximation but also preserving more favorable inductive bias than earlier kernelized methods.

Status

Research-relevant and stronger than first-wave linear attention in some settings.

5. Sparse and Structured Attention Family

These restrict token connectivity.

General form:

$$z_i = \sum_{j \in S(i)} \alpha_{ij} v_j$$

where $S(i)$ is the selected neighborhood.

5.1 Local Window Attention

$$S(i) = [i - w, i + w]$$

Key idea

Each token attends only to nearby tokens.

Use

Longformer-style models and efficient decoder designs.

5.2 Causal Sliding-Window Attention

For decoder-only models:

$$S(i) = [\max(1, i - w), i]$$

Key idea

Preserve autoregressive causality while limiting context span.

Importance

Very important in modern open LLMs.

5.3 Block Sparse Attention

Partition the sequence into blocks and attend only to selected blocks:

$$S(i) = S_{local}(i) \cup S_{global}(i) \cup S_{selected}(i)$$

Key idea

Hardware-friendlier than irregular token-level sparsity.

5.4 Longformer-Style Attention

A typical structure is:

$$S(i) = S_{window}(i) \cup S_{global}$$

Key idea

Most tokens use local windows, but designated tokens have global access.

5.5 BigBird-Style Attention

A typical pattern is:

$$S(i) = S_{local}(i) \cup S_{global}(i) \cup S_{random}(i)$$

Key idea

Combine local, global, and random edges.

Benefit

Improves graph connectivity while keeping cost subquadratic.

5.6 Top-k Attention

$$S(i) = \text{TopK}_j(s_{ij})$$

Key idea

Retain only the strongest compatibility scores.

Use

Routing-style transformers and learned sparse selection.

5.7 Local-Global Hybrid Attention

A layer-wise view is:

$$A^{(\ell)} = \begin{cases} A_{local}, & \ell \text{ odd} \\ A_{global}, & \ell \text{ even} \end{cases}$$

Key idea

Interleave local and global layers instead of mixing both in the same layer.

Importance

A very relevant 2024+ design pattern.

6. Low-Rank and Approximate Attention Family

These aim to approximate full attention more compactly.

6.1 Linformer

Project keys and values along the sequence dimension:

$$K' = EK, \quad V' = FV$$

$$A(Q, K, V) = \text{softmax}(QK^\top)V'$$

Key idea

Assume attention maps are approximately low-rank.

6.2 Nyström Attention

A simplified approximation is:

$$\text{softmax}(QK^\top) \approx F\tilde{A}^+B$$

with landmark-based factors, often written informally as:

$$A \approx QK_L^\top(K_LK_L^\top)^{-1}K_LK^\top$$

Key idea

Use landmark points to approximate the full attention matrix.

6.3 Random Feature Attention

$$\exp(q^\top k) \approx \phi(q)^\top \phi(k)$$

Key idea

Approximate softmax using positive random features.

6.4 Low-Rank Softmax Factorization

A generic form is:

$$\text{softmax}(QK^\top) \approx UV^\top$$

Key idea

Compress the full attention map itself.

7. Positional-Aware Attention Family

These modify attention by injecting relative position or geometric bias.

7.1 Relative Positional Attention

$$s_{ij} = q_i^\top k_j + q_i^\top r_{i-j}$$

Key idea

The score depends not only on content but also on relative distance.

7.2 Transformer-XL Relative Formulation

$$s_{ij} = q_i^\top k_j + q_i^\top r_{i-j} + u^\top k_j + v^\top r_{i-j}$$

Key idea

Separate content-content and content-position interactions more explicitly.

7.3 Rotary Positional Embedding (RoPE)

$$q'_i = R_i q_i, \quad k'_j = R_j k_j$$

$$s_{ij} = (q'_i)^\top k'_j$$

Key idea

Apply position-dependent rotations so inner products encode relative offsets naturally.

Importance

One of the most important positional schemes in modern LLMs.

7.4 ALiBi Attention

$$s_{ij} = q_i^\top k_j + m_h(i-j)$$

Key idea

Add a head-specific linear distance bias.

Benefit

Simple and strong for length extrapolation.

7.5 RoPE Scaling / LongRoPE / YaRN-Type Extensions

A generic view is to modify the base rotation frequencies:

$$\theta'_m = g(\theta_m)$$

Key idea

Preserve RoPE compatibility while extending context length far beyond original training length.

Importance

Critical in long-context LLM engineering.

8. Alternative Normalization and Selection Functions

These replace softmax.

8.1 Sparsemax

$$\alpha_i = \max(z_i - \tau, 0)$$

with τ chosen so $\sum_i \alpha_i = 1$.

Key idea

Project onto the probability simplex with exact zeros.

Benefit

Produces sparse attention maps.

8.2 Entmax

For $\alpha \in (1, 2]$, one form is:

$$\alpha_i = [(\alpha - 1)z_i - \tau]_+^{1/(\alpha-1)}$$

Key idea

Continuously interpolate between softmax-like and sparsemax-like behavior.

8.3 Sigmoid Attention

$$\alpha = \sigma(QK^\top)$$

Key idea

Independent gating rather than simplex-normalized competition.

Note

May require extra normalization or scaling depending on implementation.

8.4 Hard / Top-k Normalization

A conceptual form is:

$$\alpha_{ij} = \begin{cases} \frac{\exp(s_{ij})}{\sum_{j' \in S(i)} \exp(s_{ij'})}, & j \in S(i) \\ 0, & \text{otherwise} \end{cases}$$

Key idea

Selection first, normalization second.

9. Graph Attention Variants

These are domain-specific for graph-structured data.

9.1 Graph Attention Networks (GAT)

$$e_{ij} = \text{LeakyReLU}(a^\top [Wx_i \| Wx_j])$$

$$\alpha_{ij} = \text{softmax}_{j \in \mathcal{N}(i)}(e_{ij})$$

Key idea

Attention is computed only over graph neighbors.

9.2 GATv2

$$e_{ij} = a^\top \text{LeakyReLU}(W[x_i \| x_j])$$

Key idea

Improve expressiveness by changing how the linear map interacts with concatenated features.

Status

Foundational in graph learning, not a generic sequence-attention replacement.

10. Channel / Cross-Covariance Attention Family

These attend across channels rather than across tokens.

10.1 XCiT / Cross-Covariance Attention

A simplified form is:

$$A = \text{softmax}\left(\frac{K^\top Q}{\sqrt{d_h}}\right)$$

Key idea

Switch the attention axis from token-token interaction to channel-channel interaction.

Benefit

Complexity becomes much more favorable with respect to number of tokens.

Status

Primarily vision-oriented.

11. Spectral and Frequency-Domain Attention

These move attention or mixing into the frequency domain.

11.1 Spectral Attention

$$X_f = \text{FFT}(X)$$

$$A = \text{softmax}(Q_f K_f^H)$$

Key idea

Model long-range dependencies through frequency interactions.

Status

Research-oriented and modality-dependent.

12. Tensorized and Higher-Order Attention

These extend pairwise attention into higher-order interactions.

12.1 Tensorized Attention

A generic form is:

$$A_{ijkl} = q_i^\top W_{ijkl} k_j$$

Key idea

Model multi-way interactions beyond ordinary pairwise token similarity.

Tradeoff

High expressive power but heavy computational cost.

13. Memory-Augmented and Long-Context Attention

These extend effective context beyond ordinary full or local attention.

13.1 Segment Recurrence / Cached Memory

A generic recurrence is:

$$M_t = \text{concat}(\text{stopgrad}(H_{t-1}), M_{t-1})$$

Key idea

Reuse previous hidden states or memory segments.

13.2 Compressive Memory Attention

A generic form is:

$$\tilde{M}_t = C(M_t)$$

where C is a compression operator.

Key idea

Store old context in compressed form instead of keeping everything at full resolution.

13.3 Infini-Attention / Streaming Memory Attention

A conceptual decomposition is:

$$A_t = A_{local}(x_t, H_{window}) + A_{memory}(x_t, M_t)$$

Key idea

Combine local exact attention with a compressed long-term memory.

Status

Important frontier for effectively unbounded context.

14. KV-Compressed and Latent Attention Family

These are crucial in modern inference-efficient LLM design.

14.1 Multi-Head Latent Attention (MLA)

A simplified view is:

$$c_t^{KV} = W^{DKV} h_t$$

$$k_t^C = W^{UK} c_t^{KV}, \quad v_t^C = W^{UV} c_t^{KV}$$

Key idea

Compress key-value information through a latent representation rather than storing conventional full KV states per head.

Benefit

Large KV-cache reduction with strong empirical quality.

Status

Major 2024+ development.

14.2 Shared / Factorized KV State

A generic factorized form is:

$$K \approx C_K B_K, \quad V \approx C_V B_V$$

Key idea

Store and reconstruct compressed KV representations.

15. Execution-Level Attention Algorithms

These are **not new scoring rules**, but they matter enormously in practice.

15.1 FlashAttention

The output is still exact softmax attention:

$$A(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

Key idea

Reorder computation to reduce HBM traffic using tiling and online softmax.

Important note

FlashAttention is an execution algorithm, not a new attention family in the mathematical sense.

15.2 Blockwise / Ring / Distributed Attention

A conceptual partition is:

$$Q = [Q_1, \dots, Q_p], \quad K = [K_1, \dots, K_p], \quad V = [V_1, \dots, V_p]$$

Key idea

Distribute long-context attention across devices or blocks while preserving exactness or controlled approximation.

Status

Systems-critical at scale.

16. State-Space and Attention Alternatives

These are not attention mechanisms in the strict sense, but they are often compared against attention for long-sequence modeling.

16.1 Retentive Networks (RetNet)

A simplified retention update is:

$$R_t = \gamma R_{t-1} + x_t$$

$$y_t = Q_t R_t$$

Key idea

Replace full pairwise attention with a recurrent retention state.

16.2 RWKV

A conceptual weighted key-value recurrence is:

$$\text{wkV}_t = \frac{\sum_{i \leq t} e^{k_i + w(t-i)} v_i}{\sum_{i \leq t} e^{k_i + w(t-i)}}$$

Key idea

Blend RNN-style recurrence with attention-like weighted aggregation.

16.3 Mamba (Selective State Space)

$$h_{t+1} = A_t h_t + B_t x_t$$

$$y_t = C_t h_t$$

Key idea

Use data-dependent state-space parameters to model long sequences in linear time.

Important note

Mamba is **not an attention mechanism**, but it is a major long-sequence alternative.

17. Quantum / Physics-Inspired and Energy-Theoretic Variants

These are exploratory families.

17.1 Density Matrix Attention

$$\rho = \frac{XX^\top}{\text{Tr}(XX^\top)}$$

$$s_{ij} = \text{Tr}(\rho_i \rho_j)$$

Key idea

Represent token states using density-style operators.

17.2 Energy-Based Transformer View

$$E(q, k) = -\frac{q^\top k}{\sqrt{d}}$$

$$\alpha = \frac{e^{-E}}{\sum e^{-E}}$$

Key idea

Softmax attention can be interpreted as Gibbs weighting over energies.

Status

18. Unified View of Attention

Most attention mechanisms can be written as:

$$A(Q, K, V) = \text{N}(\text{S}(Q, K) + B_{pos} + M_{struct})V$$

Where:

- dot product
- bilinear
- additive
- neural energy
- kernelized
- tensorized
- graph-local similarity
- none
- relative position
- Transformer-XL bias
- RoPE
- ALiBi
- long-context RoPE scaling
- full
- causal
- local window
- block sparse
- top-k
- global-local hybrids
- softmax
- sparsemax
- entmax
- sigmoid gating
- hard top-k + local softmax

Efficiency strategies

- low-rank factorization
 - kernel linearization
 - sparse connectivity
 - KV sharing / grouping
 - latent KV compression
 - IO-aware exact execution
-

19. Big Research Insight

Modern attention research mainly improves four things.

19.1 Complexity

Target:

$$O(n^2) \rightarrow O(n), O(n \log n), \text{ or structured subquadratic}$$

Examples

- Linear Transformers
- Performer
- Nyström attention
- sparse window attention
- Mamba-style alternatives

19.2 Expressiveness

Examples

- bilinear attention
- tensor attention
- graph attention
- differential attention
- neural energy scoring

19.3 Long-Context Modeling

Examples

- RoPE
- ALiBi
- sliding-window attention
- local-global hybrids
- recurrent memory attention
- KV-compressed latent attention

19.4 Inference Efficiency

Examples

- MQA
- GQA
- MLA
- FlashAttention
- distributed blockwise attention

20. Recommended 2026 Emphasis

If the survey is intended for modern LLMs, the most important families to emphasize are:

1. Scaled dot-product attention
2. MHA / MQA / GQA
3. Differential attention
4. kernel / linear attention
5. sparse and sliding-window attention
6. local-global hybrid attention
7. Linformer / Nyström / low-rank approximations
8. RoPE and RoPE extensions
9. ALiBi
10. MLA and other KV-compressed attention
11. FlashAttention as a systems algorithm
12. memory-augmented long-context attention

The most important classification correction is this:

not every method in the literature is a direct replacement for token-token self-attention.

Some are:

- **core generic sequence attention**,
- some are **positional or structural modifiers**,
- some are **systems algorithms**,
- some are **domain-specific architectures**,
- and some are **non-attention alternatives**.

That distinction is essential for a rigorous survey.